

Trace Tables

Project: Transparent Book Recommender

1. Trace Tables

Teach the learning cycle

Primary target: Trace variable values before and after each loop iteration

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/4, ai-2/5, python-bridge/6. Reuse a familiar

case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

Look for: Trace the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks ≤ 12 only after that fails; else handles the rest. Unindented print runs after any selected branch, whose assignment gives label a value.

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: `packs = 4; per_pack = 6; total = packs * per_pack; print(total)`, each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24. Assignment alone produces no displayed output in this script.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Start `score=1`; add 4; if `score>3` subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

Concept to explain

A trace table records changing state and reveals reasoning errors.

Term · Meaning

Term: variable

Meaning: A named reference to a value that can be used or changed in a program.

Term: state

Meaning: The current values and information that determine what a process does next.

Term: trace

Meaning: A record of the successive states, decisions or information transformations in a process.

Term: iteration

Meaning: One repeat of a process, often followed by inspection and revision.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach for loops with finite integer ranges before adding more loop forms. Keep range values, body executions and output events separate. Reference: <https://docs.python.org/3/tutorial/controlflow.html#the-range-function>. The examples use a positive default step and do not require lists or while loops.

Readiness answer

It reads the old 2, adds 3 and stores 5. Assignment updates a value; it is not an algebraic equality.

Python 3 editor and loops.py

Counters or numbered paper cards

Blank trace table with turn total before and total after columns

End goal for the pictorial case

By the end, I can make a trace table for a for-loop, including zero iterations.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

2. Correct answer: $\text{score} = \text{score} + 3 \rightarrow 5$.

3. Hint used: $\text{score} = \text{score} - 1 \rightarrow 4$.

4. A table shows exactly where an unexpected value first appears.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Set $\text{score}=2$; add 3; if $\text{score}>4$ subtract 1; double score.

Trace every value.

Exact explanation to model

2,5,4,8. The comparison is evaluated at 5 before subtraction.

Explain the added worked example

Imagine adding numbered counters: first 1, then 2, then 3. A loop can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script. Tracing repeated calculations prepares you to inspect the data processing used in larger AI applications.

In for turn in `range(1, 4)`, Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body. This lesson uses integer range values with the default positive step of 1.

The output is 6. Before the first iteration, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the loop. The table records state; it does not create extra instructions.

Key terms

Term · Meaning

Term: Loop

Meaning: A block of instructions that can run repeatedly

Term: Iteration

Meaning: One execution of a loop body

Term: Accumulator

Meaning: A variable that keeps a running result

Term: Initialization

Meaning: Giving a variable its starting value

Term: Trace table

Meaning: A record of variable values as instructions execute

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

range(1, 4) · 1, 2, 3 · 3

range(1, 4): range(1, 2)

1, 2, 3: 1

3: 1

range(1, 4): range(1, 1)

1, 2, 3: No values

3: 0

range(1, 4): range(4)

1, 2, 3: 0, 1, 2, 3

3: 4

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0. Do not rely on a variable first assigned inside a loop: a zero-iteration run might never assign it.

Replace `range(1, 4)` with `range(1, 3)` in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Trace Tables

A fictional reading-token jar starts with one token and gains three per round.

The facts we will use

Python: `total=1`; for `turn in range(4)`: `total=total+3`; after the loop `print(total)`.

The update is inside the loop; print is outside.

Loop totals after each iteration

	turn	total after update
First	0	4
Second	1	7
Third	2	10
Fourth	3	13

Read a cell using both its row and column labels.

Trace Tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial state

Before the loop, total=1.



2. Iteration values

range(4) supplies turn values 0,1,2,3.



3. Updated totals

After each turn total is 4,7,10,13.



4. After the loop

Print 13 once, after all four updates.

CHANGE ONE THING

Use range(0) instead of range(4).

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Initial state

What happens and why: Before the loop, total=1.

Step: 2. Iteration values

What happens and why: range(4) supplies turn values 0,1,2,3.

Step: 3. Updated totals

What happens and why: After each turn total is 4,7,10,13.

Step: 4. After the loop

What happens and why: Print 13 once, after all four updates.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The turn value labels an iteration; it is not the number added. The fixed update adds three each time. Four iterations take the total from 1 to 13. Keeping print outside the loop produces one final display rather than four intermediate displays. Initialise total before the loop so a zero-iteration run still has a defined value.

What this example does not establish

This finite loop does not establish behavior of a while-loop with a different update or stopping condition.

Change one thing in the pictured case

Change: Use `range(0)` instead of `range(4)`.

Prediction: The loop body never runs and the printed total is 1.

Reason: The iterable contains no turn values; the initial assignment still happened.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Set `score=2`; add 3; if `score>4` subtract 1; double score.

Write the question to explain: Trace every value.

Reveal after predictions: 2,5,4,8. The comparison is evaluated at 5 before subtraction.

Keep for an independent attempt: Set score to 2. Add 3, then double the result. Record the score after

each operation. What changes if you double before adding?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

range(1, 3) supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one iteration because there are no earlier additions to lose. range(1, 4) should give 6; the faulty version gives 3.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve assignment using the prerequisite.

Stage: 7 to 20 minutes

Teacher action: Model range with numbered cards and complete W1.

Stage: 20 to 33 minutes

Teacher action: Trace each row of the worked addition and predict output.

Stage: 33 to 45 minutes

Teacher action: Complete W2 and compare predicted and observed values.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve included start and excluded stop.

Stage: 8 to 20 minutes

Teacher action: Trace the reset bug, complete W3 and discuss the guided response.

Stage: 20 to 35 minutes

Teacher action: Collect independent W4 code and zero one and several iteration traces.

Stage: 35 to 45 minutes

Teacher action: Use W5 and the exit to examine assumptions and revealing tests.

Reduce help deliberately

Completion task: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each iteration. Predict the single output, then run and compare.

Teacher check: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final output 9. Keep the observed result distinct from the prediction.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: List the supplied values before tracing; range(1, 4) excludes 4.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Move print inside the original loop. What sequence is displayed?

Reasoned teacher answer

4,7,10,13 on separate lines, because each print follows one update.

Changed-case reasoning: The loop body never runs and the printed total is 1. The iterable contains no turn values; the initial assignment still happened.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Use columns for step, event, old score, new score, and branch. Trace once by hand, predict the next row, then compare.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The stop value is included: List the supplied values before tracing; range(1, 4) excludes 4.

Initialization belongs inside every iteration: Resetting the accumulator loses earlier work.

An unindented print repeats with the body: Indentation determines which statements repeat.

Zero iterations mean the entire program does nothing: Initialization and statements after the loop still execute.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

2,5,4,8. The comparison is evaluated at 5 before subtraction.

Independent question

Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

Expected independent answer

The trace is 2, 5, 10. Doubling first gives 2, 4, 7. Operation order changes the result.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A trace table records only the final answer.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Trace is 1,5,3,6. The condition sees 5, so subtraction runs. Using the initial 1 incorrectly skips subtraction and gives 10. A variable's current state, not its first value, determines each operation.

Explanations for the additional workbook examples

W1 Explain the range

List the values supplied by `range(2, 5)` and explain why 5 is absent. How many times does its body run?
2, 3 and 4; the stop value is excluded. The body runs three times.

W2 Trace before running

Copy the worked script. Change the range to `range(2, 5)`. Record turn, total before and total after each iteration. Predict the single output, then run and compare.

Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final output 9. Keep the observed result distinct from the prediction.

W3 Repair the reset

For the faulty script with `range(1, 4)`, record total after each addition. Explain the wrong final output and remove only the faulty reset.

Faulty after-addition values are 1, 2, 3 because each starts from zero. Remove the indented `total = 0`. Correct after-addition values are 1, 3, 6; final output 6.

W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use `rounds = 3`, an initialized total and for turn in `range(rounds)`. Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Initialize `total = 5` before the loop; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final outputs for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

W5 Check a contract limitation

In your counter script, `range(-2)` performs no iterations. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

No. The written contract requires a nonnegative integer. The script assumes this and does not reject negative values. Empty range behavior does not validate the input; report the limitation or add an explicit check.

Assess independent transfer

Selected case: A box starts with 5 counters. Each round adds 2 counters. Use `rounds = 3`, an initialized total and for turn in `range(rounds)`. Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Selected case answer

Initialize `total = 5` before the loop; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final outputs for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

Trace each book's genre, length, difficulty, and total score before sorting.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 5: Loops and Stopping. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank trace grid

and permit oral explanation. Extension: put print inside the correct loop and explain why zero rounds produces no output while an outside print still does.

W4 must preserve the initial 5, update by 2, trace three iterations and print once. Require outputs 5, 7, 11 for zero one and three rounds. After support, retry with an initial 4 and additions of 3 for zero one and two rounds: outputs 4, 7, 10. Record assistance before judging independence.

Week 7 stores several values in a list and distinguishes an index from the value at that position.

Plan the delayed return

Suggested return points in this level: ai-2/5, ai-2/7, ai-2/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Trace recommender loops with current item, eligibility, score and current best so each update can be explained.

7. Prepare, teach and assess

Teaching target: Trace variable values before and after each loop iteration

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each iteration. Predict the single output, then run and compare.

Reasoned practice answer: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final output 9. Keep the observed result

distinct from the prediction.

Check every learner before discussion: Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: List the supplied values before tracing; range(1, 4) excludes 4.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use range(0) instead of range(4). Require a prediction and a reason before comparing.

Expected result: The loop body never runs and the printed total is 1. Reason: The iterable contains no turn values; the initial assignment still happened.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner’s explanation and evidence.